

Shivam Kamboj

Ph.D. Candidate in Physics | Quantum Computing | University of California, Merced

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SUMMARY

Quantum scientist (Physics Ph.D., UC Merced, anticipated Fall 2026) who builds qubits from the theory up and carries them the whole way down to hardware: **microscopic Hamiltonian** $\rightarrow$ **normal-mode and symmetry structure** $\rightarrow$ **phase diagram** $\rightarrow$ **a phase boundary exploited for device advantage** $\rightarrow$ **a fabricable circuit that realizes it** $\rightarrow$ **measurable spectrum**. My thesis found a regime of the Quantum Rabi Dimer in which parity-protected Schrödinger-cat states are the *passive* ground state of a static Hamiltonian — a biased-noise bosonic encoding needing no stabilizing drive — which I then translated into a superconducting circuit with every Hamiltonian parameter locked to a physical geometry by finite-element electromagnetic simulation. Alongside it: an open-source surface-code benchmarking library testing how fault-tolerance thresholds respond to correlated $1/f$ noise, and pulse-level control and process-tomography software delivered at LLNL for superconducting qudit hardware. First-author theory in *Physical Review B*.

TECHNICAL SKILLS

Hamiltonian & Many-Body Theory: Microscopic Hamiltonian construction and analysis; normal-mode and Bogoliubov decomposition; symmetry, parity, and conserved quantities

Device Design & Electromagnetic Modeling: Finite-element EM simulation (COMSOL RF + AC/DC); two-port S -parameter sweeps and $S \rightarrow Z$ conversion for normal-mode extraction; electrostatic energy method for femtofarad-scale capacitance; extraction of L , C , mutual inductance, parasitics and inductive energy-participation ratio; Josephson inductance and critical-current budgeting; fluxonium and lumped-LC resonator design against fabrication constraints

Quantum Error Correction: Surface-code error correction and stabilizer simulation (Stim, PyMatching), MWPM decoding, Monte-Carlo threshold estimation with bootstrap confidence intervals, correlated / non-Markovian ($1/f$) noise modeling, Ornstein-Uhlenbeck trajectory methods

Quantum Hardware & Control: Superconducting qubits/qudits (circuit-QED), photonic / cavity-QED, and magnon-photon (spin-cavity) platforms; pulse-level quantum control, gate calibration and optimization, linear-inversion quantum process tomography, qubit characterization (T1, T2, Ramsey)

Scientific Computing & HPC: Sparse linear algebra (SciPy / ARPACK, QuTiP), large-operator exponentiation, performance and memory profiling, parallel computing (multiprocessing, Joblib), SLURM scheduling, checkpointed long-running jobs at $\sim 10^4$ Hilbert-space dimension on Xeon clusters

Programming & Software Engineering: Python (NumPy, SciPy, Pandas, Matplotlib, Flask), Qiskit, Stim, PyMatching, QuTiP, TensorFlow / Keras; reproducible research engineering (pytest, ruff, mypy -strict, continuous integration); agentic development with coding agents; Git/GitHub, Linux, Bash, Google Cloud, L^AT_EX

RESEARCH & PROFESSIONAL EXPERIENCE

University of California, Merced

Merced, CA

Graduate Researcher — Tian Group (Theoretical Quantum Optics / Quantum Computing)

May 2023 – Present

- **Turned a many-body phase boundary into a qubit design target.** Identified a crossover window of the Quantum Rabi Dimer in which parity-protected Schrödinger-cat states are the *passive* ground state of a static Hamiltonian — a biased-noise bosonic encoding with structurally suppressed bit-flip errors that requires no stabilizing drive or pump, unlike dissipative and Kerr-cat approaches. Confirmed by Wigner tomography and cross-validated against independent analytical and numerical methods; first-author manuscript in preparation.
- **Translated that theory into a fabricable circuit specification.** Built a finite-element electromagnetic model (COMSOL RF and AC/DC Modules) of a superconducting layout realizing the dimer — two lumped-LC cavities, each ultrastrongly coupled to a fluxonium, joined by a flux-biased SQUID coupler — and locked every Hamiltonian parameter to a physical geometry: L_r , C_r pF, Z with inductive participation ratio extracted from the symmetric-to-antisymmetric mode ratio in a two-port S -parameter sweep. The result is a parameter-locked design ready to hand to fabrication and measurement collaborators.
- **Eliminated a design-killing parasitic before it reached fabrication**, saving a fab-and-measure cycle: electrostatic extraction showed the Josephson-array footprint alone (2.10 fF for a 20×2 μm strip) exceeded the entire 1.25 fF fluxonium island budget; compressing it to 10×1 μm brought it to 0.81 fF and made the target $E_J/E_C = 4$, $E_L/E_C = 1$ design feasible.
- **Made the underlying parameter sweeps tractable and trustworthy.** Own the Python/QuTiP codebase (sparse exact diagonalization, open-system dynamics) behind these results and scaled it into a checkpointed HPC

pipeline (SLURM, memory and worker-count tuning) at Hilbert-space dimension $\sim 10^4$; documented a trusted-regime convergence analysis that gives collaborators an explicit boundary for where the model's predictions can be relied on.

Lawrence Livermore National Laboratory (LLNL)

Physics Graduate Intern — Quantum Coherent Device Physics Group

Livermore, CA

May 2024 – Aug 2024

- **Delivered the pulse-level control software the group used to improve gate performance** on superconducting qubit hardware: designed, benchmarked, and optimized gate sequences in Python, with fidelity gains validated end-to-end by quantum process tomography rather than by proxy metrics.
- **Enabled automated device characterization** by building a generalized linear-inversion QPT pipeline (pseudo-inverse reconstruction, $\chi = \beta^+ \lambda$) that recovers process maps from noisy data, and by automating Ramsey and T1/T2 fits — giving the group rapid feedback on how measured hardware diverged from its simulated model.

Collaborator — Sharping Group (Experimental Quantum Optics), UC Merced

2024 – Present

- Serve as the modeling counterpart to an experimental group, converting their goals for a YIG-cavity magnon-photon dimer into tractable models — and showing the same Hamiltonian-level method transfers across platforms.

San Jose State University & UC Merced

Graduate Researcher — Hurst Group

San Jose / Merced, CA

Jun 2021 – Jan 2024

- Produced a first-author result on oscillatory edge modes in 2D spin-torque oscillator arrays: derived analytic dispersion relations, used them as ground truth to validate a Python solver I built, and showed the modes survive disorder. Published in *Physical Review B*.

SELECTED PROJECTS

nonmarkov_qec

Python, NumPy/SciPy, Stim, PyMatching

Open-source Python package for benchmarking QEC under non-Markovian noise — released on PyPI 2026 – Present

- **Released reusable tooling that answers a hardware-relevant question:** does temporally correlated $1/f$ noise degrade a surface-code threshold? Full pipeline — Ornstein-Uhlenbeck / sum-of-OU noise generation, Stim stabilizer simulation, MWPM/PyMatching decoding, two-layer Monte-Carlo threshold estimation with bootstrap confidence intervals — validated on rotated surface codes at $d = 3$ and $d = 5$.
- **Result de-risks a design assumption:** the threshold is invariant across three orders of magnitude in correlation time ($\tau_c = 0.2\text{--}200$ cycles, $p_{\text{th}} \approx 3.3 \times 10^{-3}$) — reported as a null result and an independent reproduction of a published finding, not dressed up as a novel effect. Shipped as production-grade software: 74 tests, `ruff` + `mypy` -strict clean, CI, automated PyPI releases.

Physics Concept Analyzer Tool (PCAT)

Python, Flask, Gemini API, GCP

Full-stack application taken from specification to production adoption at UC Merced

2025

- Built and deployed an end-user application end-to-end on Flask + Google App Engine around a structured multi-modal LLM pipeline (Google Gemini); now in active use by students and instructors.

PUBLICATIONS

1. S. Kamboj, R. A. Duine, B. Flebus, and H. M. Hurst, “Oscillatory edge modes in two-dimensional spin-torque oscillator arrays,” *Physical Review B* **109**, 094436 (2024). DOI: 10.1103/PhysRevB.109.094436
2. S. Kamboj and L. Tian, “Deterministic Cat States and Parity-Protected Bosonic Qubits in the Quantum Rabi Dimer,” manuscript in preparation, 2026.

EDUCATION

University of California, Merced

Ph.D. in Physics — Quantum Computing (Advisor: Prof. Lin Tian)

Merced, CA

Anticipated Fall 2026

Relevant courses: Computational Physics, Advanced Quantum Computing

San Jose State University (SJSU)

M.S. in Physics

San Jose, CA

2022

Relevant courses: Mathematical, Advanced Quantum Computing

AWARDS, PRESENTATIONS & OUTREACH

- **Outstanding Teaching Assistant Award** — two-time recipient: UC Merced (2026) and San Jose State University (2022). Physics Graduate Group Summer Fellowship (\$7,000/summer), UC Merced, 2025 & 2026; Travel Fellowship (\$750), 2026.
- Talks & posters: APS Global Physics Summit 2026; APS March Meeting 2022; LLNL PLS Student Poster Symposium 2024; invited seminar, CSU Stanislaus, and UC Merced PHYS 281 guest lecture, 2024.